



Thinking, Fast and Slow

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The Big “So What”

In this book, winner of the Nobel Memorial Prize in Economics, Daniel Kahneman, summarizes decades of research to help us understand what really goes on inside our heads - the psychological basis for our reactions, judgments, perceptions and choices. **By learning how our mind works, being aware of our intuitive biases and errors of judgement, we can improve our decision-making skills, shape how we think and how we live our lives.**



The Two Systems

Physicians have a large set of labels to diagnose and treat diseases (e.g. identify symptoms, possible causes, interventions and cures). Likewise, **to understand how we make judgements and choices, we need a richer vocabulary to describe how our mind works.**

Kahneman does that by simplifying complex economic and psychological concepts in this book. **He describes how our brains work, using 2 systems:**



System 1 is fast. It operates automatically and involuntarily; it is unconscious, can't be stopped, and runs continuously. We apply it effortlessly and intuitively to everyday decisions, e.g. when we drive, recall our age, or interpret someone's facial expression.



System 2 is slow. It's only called upon when necessary to reason, compute, analyze and solve problems. It confirms or corrects System 1 judgements, is more reliable, but takes time, effort

KEY QUOTES

“My main aim here is to present a view of how the mind works that draws on recent developments in cognitive and social psychology.”

“We can be blind to the obvious, and we are also blind to our blindness.”

“One of the tasks of System 2 is to overcome the impulses of System 1. In other words, System 2 is in charge of self-control.”

“The best we can do is a compromise: learn to recognize situations in which mistakes are likely and try harder to avoid significant mistakes when stakes are high.”

and concentration.

We think of ourselves with System 2 (our beliefs, choices etc.), but System 1 is usually the one in charge. System 1 operates on heuristics that are fast but often inaccurate. System 2 evaluates those heuristics and provides vigilance, but is too slow and inefficient for our routine decisions. **We need both systems, and the key is to become aware when we're prone to mistakes, so we can avoid them when the stakes are high.**



The Law of Least Effort

It takes energy to think and exercise self-control. **Our mental capacity gets depleted with use, and we're programmed to take the path of least resistance.** Since System 2 uses more effort and energy, our thoughts and actions are run automatically by System 1. **System 2 only kicks in for vital tasks that need special effort or self-control.** Specifically:

- o System 2 handles tasks that are new or unnatural to us. System 1 takes over once something's habitual, or during times of emergency.
- o System 2 focuses mental resources on the most important activity first. That's why we tend to "zone out" when we're immersed in a challenging task or are mentally overloaded, or stop in the middle of a leisurely stroll if we're asked to complete a difficult mental task.
- o We consume less energy when we're more skilled or talented at a task, can simplify the tasks into smaller steps, or write them down (to lessen our mental load).

One of the main roles of System 2 is to monitor and control the impulses and suggestions by System 1. But, **System 2 fails to perform its role when it's lazy or overloaded.** That's why most people don't think through problems carefully, and why we find it harder to resist temptation when we are stressed. In short, **self-control decreases when we're tired, hungry, or mentally exhausted.**

KEY QUOTES

"In the economy of action, effort is a cost, and the acquisition of skill is driven by the balance of benefits and costs. Laziness is built deep into our nature."

"We cover long distances by taking our time and conduct our mental lives by the law of least effort"

"For most of us, most of the time, the maintenance of a coherent train of thought and the occasional engagement in effortful thinking also require self-control."

"Intense focus on a task can make people effectively blind, even to stimuli that normally attract attention."



Heuristics and How System 1 Works

In order to quickly process the tons of stimuli that we take in daily, System 1 uses “heuristics” or mental shortcuts. These are fast, but often unreliable, and makes us susceptible to others’ suggestions or influence.



Associations and Priming. When consciously or subconsciously exposed to an idea, we’re “primed” to think about associated idea(s), memories & feelings. If we’ve been talking about food, we’ll fill in the blank SO_P with a “U”, but if we’ve been talking about cleanliness, we’ll fill in the same blank with an “A”. People reading about the elderly unconsciously walk slower, and people who are asked to smile find jokes funnier. Each associated idea evokes even more ideas, memories, and feelings. This is called the associative mechanism.



Cognitive Ease. System 1 is constantly assessing stimuli and impressions, to determine if System 2 needs to be mobilized:

- o When we experience “cognitive ease”, we’re in a casual, intuitive mode. System 1 is at work and we don’t think deeply. Generally, we feel at ease when something seems familiar, is clear and effortless. That’s why politicians and marketers use repetition, bright colours, legible words, simple language and rhyming verses. We’ll even believe a lie if we hear it often enough.
- o When we experience “cognitive strain”, System 2 kicks in and we switch to an analytical, problem solving mode. We become more vigilant and are less susceptible to lies and suggestions.



Stories and Causes. To make sense of the world, we connect scraps of information and tell ourselves stories about what’s going on.

- o Associative Coherence. We make associations between events and circumstances. Events that fit

KEY QUOTES

“System 1 is gullible and biased to believe, System 2 is in charge of doubting and unbelieving, but System 2 is sometimes busy, and often lazy.”

“You know far less about yourself than you feel you do.”

“Like ripples on a pond, activation spreads through a small part of the vast network of associated ideas.”

“Anything that makes it easier for the associative machine to run smoothly will also bias beliefs.”

“Cognitive strain, whatever its source, mobilizes System 2 which is more likely to reject the intuitive answer suggested by System 1.”

“People are prone to apply causal thinking inappropriately, to situations that require statistical reasoning.”

into our stories seem “normal”. When something unexpected happens, we tell ourselves new stories to make them fit e.g. “It’s God’s will,” or “He acted out of character.”

- o **Jumping To Conclusions.** We often have to deal with incomplete information. System 1 fills the gaps with interpretations and guesses that fit our stories, jumping to conclusion on causes.
 - i. **Confirmation Bias** is our tendency to search for and find confirming evidence for a belief (while dismissing counter evidence). During an election season, we’re likely to seek positive information about our favored candidates, and find negative information about the opposing candidate(s). System 1 also usually flags out only one conclusion and omits alternatives.
 - ii. **The Halo Effect** occurs when we like or dislike everything about a person, place, or thing, based on our first impressions. The first person to share an opinion at a meeting can “prime” others’ opinions. Positive adjectives describing a person (e.g. strong, intelligent) influence how we interpret negative adjectives (e.g. stubborn, impulsive) that appear later.
 - iii. **What You See is All There Is (WYSIATI).** System 1 ignores whether the information used is complete or accurate. It only cares how coherent the story is, based on available information.



Judgement Heuristic. Unlike System 2, which zooms in to find specific answers, **System 1 adopts a “Mental Shotgun” approach**, continuously scanning our environment and making “fight or flight” judgements.

When making such split-second decisions, **we’re better at broad averages and categorical comparisons, than sum-like variables.** For instance, we can intuitively describe crime in terms of colours (“murder” would be a deeper shade of red than “theft”). We are also better at **intuitive averages** than sum-like variables (we can understand “about twice the current size” more easily

KEY QUOTES

“We have limited information about what happened on a day, and System 1 is adept at finding a coherent causal story that links the fragments of knowledge at its disposal.”

“When information is scarce, which is a common occurrence, System 1 operates as a machine for jumping to conclusions”

“System 1 does not keep track of alternatives that it rejects, or even of the fact that there were alternatives.”

“Sequence matters... because the halo effect increases the weight of first impressions, sometimes to the point that subsequent information is mostly wasted.”

“It is the consistency of the information that matters for a good story, not its completeness.”

than “an increase from 20,000 to 40,000”).



Substitution. When faced with a vexing problem or question, *we make our lives easier by answering a simpler, substitute question*. For example, “What’s my mood right now?” is easier to answer than “How happy am I with life?”. This sometimes means that we avoid answering the harder (but more important) questions.



Affect or Emotions. *When we have a strong emotional preference for an outcome, we will play up the benefits and play down its costs*. In addition, a simpler question that evokes an emotional response will influence our response to a bigger question. If we are asked “how many dates did you have last month?” just before we’re asked “how happy are you with your life?”, we’ll link our happiness to the number of dates, even if dates weren’t important to us.

In short, *System 1 works using shortcuts* like associations, stories and approximates, tends to confuse causality with correlation, and jumps to inaccurate conclusions. *System 2 is supposed to be our inner sceptic, to evaluate and validate System 1’s impulses and suggestions*. But, it’s often too *overloaded or lazy to do so*. This results in intuitive biases and errors in our judgement.



Heuristics Cause Biases & Errors



The Law Of Small Numbers

Most of us know that small sample sizes are not as representative as large samples. Yet, System 1 intuitively believes small sample outcomes without validation. We make decisions based on insufficient or unrepresentative data. Moreover, System 1 suppresses doubt by constructing coherent stories and attributing causality. Once a wrong conclusion is accepted as “true”, it triggers our associative mechanism, spreading related ideas through our belief systems.

KEY QUOTES

“You often have answers to questions that you do not completely understand, relying on evidence that you can neither explain nor defend... you may not even notice that you did not answer the question you were asked.”

“People let their likes and dislikes determine their beliefs about the world.”

“A lazy System 2 often follows the path of least effort and endorses a heuristic answer without much scrutiny of whether it is truly appropriate.”

“System 1 is designed to jump to conclusions from little evidence – and it is not designed to know the size of its jumps.”

“The law of small numbers is a manifestation of a general bias that favors certainty over doubt.”



The Anchoring Effect

Our estimates or choices are influenced (subconsciously) by number(s) that we heard previously. For instance, your answer to “Was Gandhi more than 116 years old when he died?” is likely to be closer to 116 than 50. Shoppers are likely to buy more of a promotional item if there was a “limit of 12 per person”, than if there was no limit.



The Availability Heuristic

We tend to judge the importance of an idea by the familiarity and feelings that we associate with it, rather than statistical calculation. **We are prone to overestimate the frequency of an event if we are fluent with the idea**, e.g. we may overestimate the frequency of divorces if we have undergone divorce, or know people who're divorced.

This heuristic is multiplied via media coverage to create “Availability Cascades”, and have implications on public policy.

- o **Media coverage warps our estimates of events**. A recent plane crash makes us think air travel is dangerous. The more we fear air travel, the more the media report plane crashes. This creates a negative feedback loop or a cascade of fear, and may trigger a surge in insurance purchases or protective action after a disaster.
- o Likewise, **public policy can be influenced by such irrational concerns** - lawmakers may be prone to the same cognitive biases as citizens, or respond due to political sensitivity.



Representativeness

We tend to make judgments about a person, place, or thing based on how much it resembles something else. We may stereotype that someone who's “nerdy” is probably a computer scientist, or assume that a restaurant is doing well because we like their food. A more accurate approach would be to look at probability, base rates, and examine the evidence that was used to develop our assumptions.

KEY QUOTES

“The psychological mechanisms that produce anchoring make us far more suggestible than most of us would want to be.”

“The main moral of priming research is that our thoughts and our behaviour are influenced, much more than we know or want, by the environment of the moment.”

“The dynamics of memory help explain the recurrent cycles of disaster, concern, and growing complacency that are familiar to students of large-scale emergencies.”

*“The emotional tail wags the rational dog.”
- Jonathan Haidt*

“Stereotypes, both correct and false, are how we think of categories.”



The Conjunction Fallacy (Less is More)

This occurs when we assume that A+B is more probable than A, e.g. assuming that this statement: “He has lived here his whole life and knows the culture well” is more probable than “He has lived here his whole life.” This is flawed because A+B is a subset of A, and can only be less probable.

Likewise, we are likely to perceive an expensive item to be more valuable on its own than when it's bundled with additional cheap items. System 1 intuitively processes averages better than sums.



Causes Trump Statistics

Statistical data are facts about a case, e.g. “50% of cabs are blue”. Causal data are facts that change our view of a case, e.g. “blue cabs are involved in 80% of road accidents” - we may infer from the latter that blue cab drivers are more reckless.

We've learnt that System 1 thinks fast using categories and stereotypes, and likes causal explanations. When we're given statistical data and causal data, we tend to focus on the causal data and neglect or even ignore the statistical data. In short, we favor stories with explanatory power, over mere data.



Regression To The Mean (Overlooking luck)

Random fluctuations (e.g. an unusually good or bad sports performance) will typically regress to the mean (i.e. they'll naturally approach the average over time). We tend to assign causal explanations to this phenomenon when they don't exist. We may punish someone for unusually bad behaviour, then convince ourselves that the punishment was effective when the person merely regressed naturally to his normal behaviour. The proper way to establish causal relationships is through control-group experiments.

KEY QUOTES

“Sometimes intuition beats logic even when the correct answer stares you in the face.”

“The notions of coherence, plausibility, and probability are easily confused by the unwary.”

“Our mind is strongly biased toward causal explanations and does not deal well with ‘mere statistics’.”

“You are more likely to learn something by finding surprises in your own behaviour than by hearing surprising facts about people in general.”

“The truth is regression to the mean has an explanation but does not have a cause.”



Heuristics Cause Overconfidence

We feel confident when our stories seem coherent and we are at cognitive ease. Unfortunately, confidence does not mean accuracy.



The Illusion Of Understanding

- **The Narrative Fallacy.** We often create flawed stories to explain the past, which in turn shape our views of the world and our expectations of the future. For example, we may develop detailed models and formulas to explain what caused the global financial crises or Google's rise to success. But it's impossible to identify all the variables and events that happened (and those that didn't happen) to cause these outcomes. No one could have predicted the birth of Adolf Hitler and how it catalyzed World War II. So, we understand the past less than we think we do; Believing we know the past and can predict the future gives us a false sense of security.
- **The Hindsight Illusion.** Once we change our view of the world, we immediately lose much of our ability to recall what we used to believe. We feel as though our current belief has been our belief all along. This leads to hindsight bias (the "I-knew-it-all-along" effect). That's why it's easy to blame others for "bad" decisions on hindsight and give too little credit for good decisions.



The Illusion Of Validity

- **Skills Illusion.** We are prone to think that skills produce predictable results, e.g. experienced traders are believed to be better at evaluating and picking stocks. However, a truly efficient market cannot be predicted nor outsmarted, and studies have shown that **traders' "educated guesses" are no better than blind guesses.**
- **Formulas vs Intuitions.** **Generally, statistical algorithms are better predictors than experts,** because experts often try to "think out the box" and consider complex combinations. Their creative solutions sometimes work,

KEY QUOTES

"Confidence is a feeling, which reflects the coherence of the information and the cognitive ease of processing it."

"We humans constantly fool ourselves by constructing flimsy accounts of the past and believing they are true."

"The ultimate test of an explanation is whether it would have made the event predictable in advance."

"You cannot help dealing with the limited information you have as if it were all there is to know. You build the best possible story from the information available to you, and if it is a good story, you believe it."

"Actions that seemed prudent in foresight can look irresponsibly negligent in hindsight."

but reduce their overall reliability. Humans are also inconsistent - the same person can give different answers when asked to judge the same complex situation twice.

- **Trusting Expert Intuition.** Expert intuition can still be trustworthy and valuable under these conditions:
 - o If the environment is regular enough to be predictable; and
 - o The expert has had a chance to learn these regularities via prolonged practice (with timely, quality feedback)

Some examples include chess-games, fire-fighting, and nursing.



The Optimistic Bias

Optimism is a positive trait. However, **overconfidence** and the **optimistic bias** can also lead people to take excessive risks.

- **Planning fallacy** is one outcome of this bias - **people take on risky projects** (e.g. litigation or starting a business) **based on best case scenarios**, without deliberating worst case scenarios or unknowns.
- **Outside View.** To address this fallacy, planners could adopt **a more objective "outside view"**. Identify others who've engaged in similar projects, use their statistics as a baseline, then adjust the baseline using information specific to your case.



Heuristics affect Choices



Prospect Theory

This theory won Kahneman the Nobel Prize in Economics. It's built on 3 key ideas:

KEY QUOTES

"Valid intuitions develop when experts have learned to recognize familiar elements in a new situation and to act in a manner that is appropriate to it... Unfortunately, professionals' intuitions do not all arise from true expertise."

"Expertise...is a collection of skills, and the same professional may be highly expert in some of the tasks in her domain while remaining a novice in others."

"Intuition cannot be trusted in the absence of stable regularities in the environment."

"In terms of its consequences for decisions, the optimistic bias may well be the most significant of the cognitive biases."

"Most of us view the world as more benign than it really is, our own attributes as more favorable than they truly are, and the goals we adopt as more achievable than they are likely to be."

- The absolute value of money is less vital than the **subjective experience** that comes with changes to your level of wealth. For example, having \$5,000 today is “bad” for Person A if he owned \$10,000 yesterday, but it’s “good” for Person B if he only owned \$1,000. The same \$5,000 is valued differently because **people don’t just attach value to wealth – they attach values to gains and losses.**
- We experience **reduced sensitivity to changes in wealth**, e.g. losing \$100 hurts more if you start with \$200 than if you start with \$1000.
- **Loss aversion.** **We basically hate to lose money, and we weigh losses more than gains.** To illustrate, look at these 2 options:
 - o 50% chance to win \$1,000 OR get \$500 for sure
Most people will choose the latter
 - o 50% chance to lose \$1,000 OR lose \$500 for sure
Most people will choose the former

Generally, our brain processes threats and bad news faster, people work harder to avoid losses than to attain gains, and they work harder to avoid pain than to achieve pleasure.



The Endowment Effect

An object that we own and use (or intend to use) is more **valuable to us**. We endow such objects with additional value, and we’re unwilling to part with them. For example, you may buy a ticket for your favourite concert for \$500, and refuse to sell it even for \$5,000.



The Fourfold Pattern of Preferences

The table on the next page summarizes how we evaluate gains and losses.

KEY QUOTES

“You just like winning and dislike losing—and you almost certainly dislike losing more than you like winning.”

“We are driven more strongly to avoid losses than to achieve gains.”

	Gains	Losses
HIGH PROBABILITY (Certainty Effect)	95% chance to win \$10,000 Risk Averse (e.g. court settlements) 1	95% chance to lose \$10,000 Risk Seeking (e.g. desperate gambles) 2
LOW PROBABILITY (Possibility Effect)	5% chance to win \$10,000 Risk Seeking (e.g. lottery tickets) 3	5% chance to lose \$10,000 Risk Averse (e.g. insurance) 4

- **Certainty Effect:** People are generally risk-averse when they have a high chance of getting a desired outcome.

- In Quadrant 1 (high probability, big gains), people will accept a less-than-expected value of the gamble to lock in a big gain. For example, someone may accept a \$910,000 settlement rather than go for a lawsuit with a 95% chance of winning \$1 million (i.e. they accept \$40,000 less than the expected gamble value).
- In Quadrant 4 (low probability, big loss), people are willing to pay a premium for the certainty of not losing. They pay more to buy a peace of mind (e.g. insurance).

- **Possibility Effect:** We tend to give an irrationally high weightage to a desired but improbable outcome:

- In Quadrant 3 (low probability, big gains), people over-invest in a miniscule chance to win (e.g. lottery tickets)
- In Quadrant 2 (high probability, big losses), people take desperate gambles, often making things worse in exchange for a small hope of avoiding a large loss. This is why people throw in everything they have to treat a terminal illness, or sink in all their business assets in a futile attempt to catch up with competition.

When we combine the Possibility Effect, Availability Cascade and Cognitive Ease heuristics, we start to understand why **people pay an undue amount of attention to rare events (e.g. terrorist attacks, asteroids)**, compared to likely events (e.g. rain tomorrow). This is especially so when the rare events are described vividly, frequently and repetitively.

Earlier, we shared how to adopt an “outside view” to address planning fallacies from excessive optimism. **To counter the excessive caution from loss aversion, we can develop “risk policies”** (e.g. never buy extended warranties), which act as broad frames to guide our routine decisions.

KEY QUOTES

“When you pay attention to a threat, you worry – and the decision weights reflect how much you worry. Because of the possibility effect, the worry is not proportional to the probability of the threat.”

“When you take a long term view... you can see that paying a premium to avoid a small risk of a large loss is costly... Consistent overweighting of improbable outcomes – a feature of intuitive decision making – eventually leads to inferior outcomes.”

“The outside view and the risk policy are remedies against two distinct biases that affect many decisions: the exaggerated optimism of the planning fallacy and the exaggerated caution induced by loss aversion.”



Keeping Score

Except for the very poor, most money-seekers aren't just after economic gains - money also symbolizes success and achievement. Hence, people keep score of both (a) the potential financial gains and losses of a transaction, and (b) the emotional risks, rewards, and possible regrets of these decisions:

- The **Disposition Effect** refers to our tendency to sell "winners" (stocks that currently have a higher price than when purchased) rather than "losers" (stocks that're still at a loss). This is an example of "narrow thinking."
- The **Sunk-cost Fallacy** is the tendency to invest additional resources in a losing account, instead of other better investments. Cutting loss is emotionally difficult as it signifies failure. By gambling further, the decision-maker hopes to recoup the original investment or at least postpone the day of reckoning. This fallacy keeps people in poor jobs, unhappy marriages, and unpromising projects.
- **Fear of regret** influences many of our decisions. Consumers who are reminded of potential regret tend to choose safer, conventional options. Don't put too much focus on regret; it usually hurts less than you expect.



Joint Evaluations

We make decisions differently when they're done in isolation than when they are made in comparison with other scenarios. For example, we pay more for a "better-quality" TV that's displayed next to an "inferior" TV in the store, even when the difference in quality is imperceivable when the TVs are viewed in isolation.



Frames

How a problem is framed evokes emotional responses, which affect our choices. Because we hate to lose, we'll intuitively choose a "10% chance to Win" over a "90%

KEY QUOTES

"The emotions that people attach to the state of their mental accounts are not acknowledged in standard economic theory."

"A rational decision maker is interested only in the future consequences of current investments."

"Regret is an emotion, and it is also a punishment that we administer to ourselves."

"Joint evaluations highlights a feature that was not noticeable in single evaluations but is recognized as a decisive when detected."

"The meaning of a sentence is what happens in your associative machinery while you understand it."

chance to Lose”. Doctors prefer interventions if there’s a “survival rate of 90%” vs a “10% mortality rate.” A closer look in these cases will reveal that the options are actually the same; only the frames are different.



Two Selves

We each have an “experiencing self” and a “remembering self”. In his research on happiness, Kahneman found that our memories override our actual experiences, and we make decisions with the aim of creating better memories (not better experiences).

“ ” **The Peak-End Rule.** We evaluate our life like a story, focusing on significant events, memorable moments, and how it ends. **The actual experience is not as important as the memory of how painful or pleasurable the experience was**; our remembering self makes decisions based on the memories we expect to create. In fact, during an experiment, people chose to repeat a painful experience that lasted longer but ended pleasantly, over a shorter painful experience without a pleasant ending.

👤 **Experienced Well Being.** When we resist ending an experience, it means we’re having a good time. To improve our well-being, we should spend time on things that we enjoy and pay attention to what we are doing, so we get more pleasure from our current activities and create positive memories of these experiences.



Thinking About Life

- Generally, we’re terrible at “**affective forecasting**”, or predicting what will make us happy. People make decisions based on what they think will make them happy in the future, but find out when they “get there” that the happiness doesn’t last, e.g. the satisfaction of marriage drops dramatically after the initial years.

KEY QUOTES

“Reframing is effortful and System 2 is lazy.”

“Confusing experience with the memory of it is a compelling cognitive illusion.”

“A story is about significant events and memorable moments, not about time passing... the ending often defines its character.”

“The remembering self... composes stories and keeps them for future references.”

“We all care intensely for the narrative of our own life and very much want it to be a good story, with a decent hero.”

“Nothing in life is as important as it is when you are thinking about it.”

- Our levels of happiness and life satisfaction depend on many factors. When we focus on only one area (e.g. income, health, relationships), we suffer from the **Focusing Illusion** and cannot assess our happiness accurately. We are also prone to acclimatization – what’s initially exciting (e.g. a new car) loses appeal over time.



Conclusion

Our heuristics influence our choices, which can be irrational, counter-intuitive and sub-optimal. It’s impossible to totally avoid biases and errors from System 1, but we can make a deliberate effort to slow down and utilize System 2 more effectively, especially when stakes are high.

KEY QUOTES

“This book has described the workings of the mind as an uneasy interaction between two fictitious characters: the automatic System 1 and the effortful System 2.”

“As we have seen again and again, an important choice is controlled by an utterly inconsequential feature of the situation This is embarrassing – it is not how we would wish to make important decisions.”

“The way to block errors that originate in System 1 is simple in principle: recognize the signs that you are in a cognitive minefield, slow down, and ask for reinforcement from System 2.”